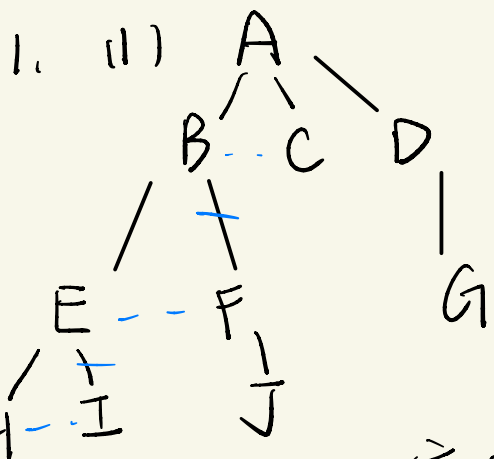


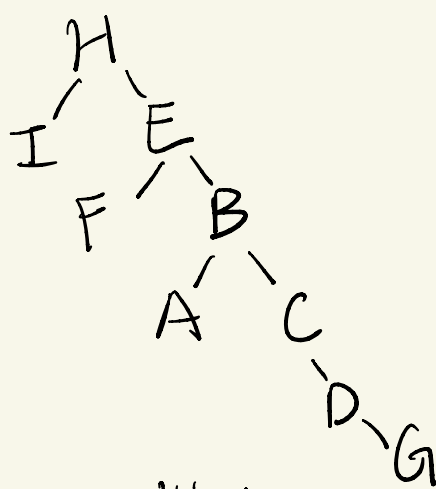
六



$(A(B(E(H, I), F(J)), C, D(G)))$

(2) 深4 高3

(3)



```

if kNode.child is not empty:
    Node.left = convert(Node.child)
    pos = Node.right
    for i from 1 to length:
        right = convert(Node)
        pos = pos.right
    
```

2. (1) $\left[\frac{k^k - 1}{k - 1}, \frac{k^{k+1} - 1}{k - 1} \right]$

(2) $\left\lfloor \frac{N-1}{k} \right\rfloor$

(3) $k \cdot N + i + 1$

3. 证明: 当 $n=1$ 时, 深度为0, 满足

假设 $n=k$ 时成立

$n=k+1$ 时 合并两集合时, 假设高度分别为 h_1, h_2

且 $h_1 \leq h_2$ $\begin{cases} \text{若 } h_1 < h_2 & \text{合并后树高仍为 } h_2 \\ h_1 = h_2 & \text{合并后树高 } h_2 + 1 \end{cases}$

则每次 union

树高最多+1, 则进行 $k+1$ 次

树高最大为 $\log_2(k+1)$ 证